

RV32I Monocycle Processor Verification-Oriented Specifications

32-bit RISC-V

Draft

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January 22, 2026

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Glossary

- **RV32I** — Base integer instruction set for 32-bit RISC-V.
- **ISA** — Instruction Set Architecture.
- **PC (Program Counter)** — Byte-addressed register holding the address of the current instruction.
- **Monocycle / Single-cycle** — Each instruction is fetched, decoded, executed, and retired in one clock cycle.
- **Register File** — Architectural register set **x0–x31**, 32-bit each.
- **x0 (Zero Register)** — Architectural register hardwired to zero; reads return 0 and writes have no effect.
- **Immediate (imm)** — Constant encoded in the instruction and sign-extended to 32 bits.
- **funct3/funct7** — Instruction subfields used to select the specific operation within an opcode.
- **Alignment** — Word alignment implies $\text{PC}[1:0] = 2'b00$; misaligned accesses are out of scope.
- **DUT** — Device Under Test.

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1 Introduction

This document is a concise, verification-oriented specification for a 32-bit RV32I monocycle processor. It is intended to:

- Define the supported instruction classes (R, I, B, J) and their architectural semantics.
- Describe instruction structure (fields, immediate encoding) for reference modeling.

Out of scope: privileged architecture, exceptions, interrupts, CSR behavior, misaligned accesses, compressed (C) extension, multiplication/division (M), atomics (A), floating point (F/D), and any multi-cycle/pipelined behavior.

2 RV32I Monocycle Processor

2.1 Overview

The processor implements the RV32I base ISA and executes one instruction per clock cycle (monocycle). The architectural state comprises the program counter (PC) and the integer register file. Memory is modeled as instruction memory (read-only) and data memory (loads/stores).

2.2 Architectural Assumptions

- Byte-addressed PC. Draft
- No stalls, no pipeline, no flush, no speculation.
- No exceptions/interrupts; illegal instructions are not exercised by verification.
- Load/store addresses are assumed aligned for word accesses.

2.3 Instruction Classes Supported

Class	Format	Supported Operations (minimum set for verification)
R-type	R	ADD, SUB, AND, OR, XOR, SLT
I-type	I	ADDI, LW
B-type	B	BEQ
J-type	J	JAL

2.4 Top-Level Port Conventions (DUT Interface)

Port	Bits Used	Type	Description
clk	[0]	Input	Global clock. One instruction retires per rising edge.
arst_n	[0]	Input	Active-low async reset (if used by your RTL).
instr	[31:0]	Input	Current fetched instruction.

3 Instruction Structure

3.1 Common Fields

Field	Bits	Meaning
opcode	[6:0]	Major instruction class selector.
rd	[11:7]	Destination register index.
funct3	[14:12]	Sub-operation selector within an opcode.
rs1	[19:15]	Source register 1 index.
rs2	[24:20]	Source register 2 index.
funct7	[31:25]	Extended sub-operation selector for R-type operations.

3.2 R-Type Format

funct7[31:25]	rs2[24:20]	rs1[19:15]	funct3[14:12]	rd[11:7]	opcode[6:0]
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3.3 I-Type Format

imm[11:0][31:20]	rs1[19:15]	funct3[14:12]	rd[11:7]	opcode[6:0]
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3.4 B-Type Format (Branch Immediate)

imm[12 10:5]	rs2	rs1	funct3	imm[4:1 11]	opcode
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3.5 J-Type Format (Jump Immediate)

imm[20 10:1 11 19:12]	rd[11:7]	opcode[6:0]
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4 Operation Semantics (Architectural)

Figure 1 illustrates the architectural datapath of the RV32I single-cycle microprocessor implemented in this work. The design follows a classical single-cycle organization, where instruction fetch, decode, execution, memory access, and write-back occur within one clock period.

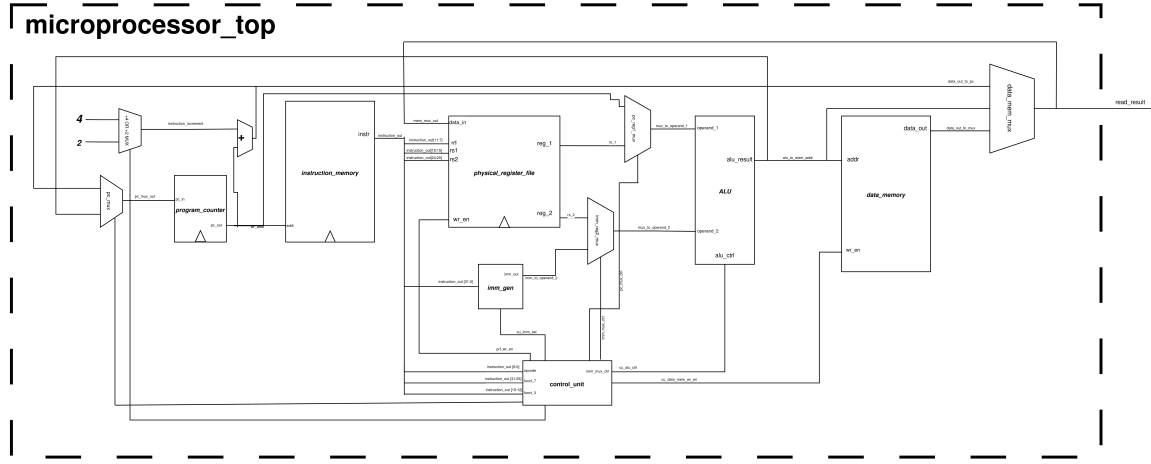


Figure 1: Architectural datapath of the RV32I single-cycle microprocessor.
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Program Counter (PC)

The program counter stores the address of the current instruction. Its next value is selected through a multiplexer that allows sequential execution ($PC+4$ or $PC+2$) or control-flow redirection for branch and jump instructions.

Instruction Memory

The instruction memory is a read-only memory indexed by the word-aligned program counter. Instructions are fetched using a byte-addressed PC, with the two least significant bits discarded to obtain a word index.

Control Unit

The control unit decodes the instruction opcode and function fields and generates all control signals required by the datapath, including register write enable, ALU operation selection, operand multiplexing, memory access control, and write-back source selection.

Immediate Generator

The immediate generator extracts and sign-extends immediate fields according to the instruction format (I, S, B, J, or U), providing properly formatted constants to the ALU.

Physical Register File

The register file provides two read ports and one write port. Register x0 is hardwired to zero, and write attempts to this register are ignored in accordance with the RISC-V architectural specification.

Operand Selection Multiplexers

Operand multiplexers select between register values, immediate values, or the program counter as inputs to the ALU, enabling arithmetic, logical, and control-flow operations.

Arithmetic Logic Unit (ALU)

The ALU performs arithmetic and logical operations as specified by the instruction. It also produces comparison results used for branch decision logic.

Data Memory

The data memory is accessed during load and store instructions. The address is provided by the ALU, while write enable and data routing are controlled by the control unit.

Write-Back Multiplexer

The write-back multiplexer selects the value written to the register file, choosing between the ALU result, data memory output, or immediate/PC-related values depending on the instruction type.

5 References

- RISC-V International, *The RISC-V Instruction Set Manual, Volume I: Unprivileged ISA*, Version 20191213.